David Pérez Camacho

Mobile Tech Lead

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blueheisenberg.com scskiller.com

Android · iOS · Flutter · KMP | BLE & connected hardware | Custom Android / AOSP | Embedded firmware | AI orchestration & local LLMs

Mobile engineer and tech lead with **15+ years** across **Android, iOS and Flutter**, specialising in apps that connect to hardware over **BLE**. **Mobile Tech Lead at Stark Future**, where I also lead **AI and agentic workflow** adoption across engineering. Outside work I build with my own **Claude Code orchestration system**, including an open-source app with **100k+ users**. I work across the whole stack, from firmware and custom Android to backend and CI/CD.

15+ years in software	**100k+** open-source app users	**4k+** GitHub stars	**>10,000** bikes running our app

AI-ASSISTED ENGINEERING

Claude Code orchestration
Built my own **multi-agent orchestration system** on Claude Code, coordinating agents across the development workflow. Built and shipped **SCSKiller** end to end with it.

Local multi-GPU AI
Run a **multi-GPU local AI setup** (vLLM, llama.cpp, quantized models) wired into the orchestration, so local models share the work with Claude and power product features.

Team adoption
At Stark Future, drove **company-wide adoption** of AI-assisted development (Claude Code, Cursor) and built the **shared skills and automations** teams use daily.

OPEN SOURCE

SCSKiller · scskiller.com github.com/BlueHeisenberg/SCSKiller

Open-source Windows app that precompiles DirectX 12 shaders before you play, so games don't stutter. **4k+ GitHub stars** and **100k+ users**.

- Built entirely with my **Claude Code orchestration system**, from the shader pipeline to the website.
- Telemetry and backends for all those users run on a single small VPS for about **€10 a month**.
- **Community support on Discord** with built-in assistance from my **AI orchestration system**, including its local models.

EXPERIENCE

Mobile Tech Lead · Stark Future CURRENT — Dec 2023 – Present

Mobile ecosystem for an electric motocross platform: the bike's dashboard app, a custom Android phone that ships with the bike, and a new rider companion app.

- Grew the mobile team **from 3 to 7** and split it into dashboard and companion app teams; product owner for mobile.
- Architect and maintain the **Stark VARG Dashboard app**, running on **tens of thousands of bikes** worldwide, with a **very high-throughput BLE** connection for live data and telemetry uploaded to the cloud, and riders' watches connected for heart rate. Rebuilt it from scratch in **Jetpack Compose** (Coroutines, Retrofit, Room, Koin) while keeping the legacy app running.
- Built multiple **Flutter** apps that connect to the VARG bike over **BLE** and **Wi-Fi**.
- Building a cross-platform companion app with **Kotlin Multiplatform** and **Compose Multiplatform**, sharing UI and logic between iOS and Android; created an internal **NFC** tool for reading bike VINs.
- Own the **custom phone project** end to end: system-level app, **AOSP** and firmware changes, OTA pipeline and manufacturer coordination.
- Designed secure bike-to-phone **BLE connectivity**, with BLE optimizations alongside the embedded team.
- Contributed to the bike's **embedded firmware** across its multi-chip architecture and microcontroller optimizations.
- **CI/CD** with GitHub Actions (Google Play and AWS internal distribution) and Jenkins for firmware builds; AWS Lambda services for mobile backends.

Chief Technology Officer · NEN (health-tech)

Dec 2022 – Dec 2023

Designed and built the company's tech stack alone for the first months, then managed one engineer.

- Built **two Flutter apps** for iOS and Android, one for parents and one for kids, on a serverless **Firebase** backend (Firestore, Storage, Auth, Cloud Functions), free during clinical trials and the first release.
- Built a content pipeline with **AI in the loop**: a terminal app to add text, Cloud Functions for translation into 7 languages (**OpenAI API**) and voiceovers (**ElevenLabs**), and a CMS-style website to validate it for production.
- Led technical **GDPR and HIPAA** compliance and set up CI/CD and the company's IT processes.

CTO & Head of Hardware Development Operations · Smiletronix

Aug 2019 – Dec 2022

Joined as Head of Mobile and moved on to lead embedded and hardware development for the company's device.

- Managed **three staff** and, with freelancers, up to **nine people**, plus external partners across hardware, firmware and apps.
- Designed and wrote the **embedded software**, on a custom **Yocto Linux**, for an in-mouth device with **nine cameras**, which streamed images to the phone over **Wi-Fi**, with **BLE** for setup and control.
- Built the **iOS (Swift, SwiftUI)** and **Android (Kotlin)** apps, CI/CD with Bitrise and Bitbucket Pipelines, a custom **Python** backend, and AI inference on GCP, AWS and Azure.

Freelance Engineer

Jun 2015 – Present

- Designed and built a portable **BLE electro-muscular stimulator** for therapy: PCB design, **Nordic nRF51** firmware and iOS/Android apps.
- Ported a web app to native Android with BLE 4.1 and real-time **C++ through JNI/NDK**.
- **Discovery Channel** Android app; Node.js backend for an AXA-group app.
- Games: ported **The Crow's Eye** (Unity) to macOS and Linux; optimization and DevOps on **Arima** (Unreal Engine 4, Switch/PC).
- Built Rapit, a QR-code restaurant menu platform.

EARLIER CAREER

Senior Android Developer, Future Workshops — Riyadh Airports app	2017–2018
Senior Main Android Developer, Monqi — led the Android team; MQTT over AWS IoT	2017
Senior Android Developer, Digital Origin — led Android; Quebueno.es, PagaMasTarde.com, NFC payment pilot	2015–2017
Lead Android Developer, Toro Development — NFC payments wallet; led two engineers	2014–2015
Senior Android Developer, Altran — AXA apps, Samsung Gear (Tizen)	2014
CTO & Chief Android Developer, Devinet — Allianz app with audio/image recognition	2013–2014
Android & Web Developer, V2MSoft, XAL and others — incl. a Bluetooth tool for Manusa doors	2010–2013

SKILLS

Mobile	Kotlin, Jetpack Compose, Kotlin & Compose Multiplatform, Swift / SwiftUI, Flutter / Dart, Coroutines, Ktor, Clean Architecture, MVVM, Koin / Dagger, Room, Retrofit
Hardware & embedded	BLE, NFC, Wi-Fi device streaming, AOSP and custom firmware, Yocto Linux, OTA, Nordic nRF, microcontrollers, C / C++, JNI / NDK, electronics and PCB design
Cloud & DevOps	Firebase, AWS (Lambda, IoT), GCP, Azure, GitHub Actions, Jenkins, Bitrise; GDPR / HIPAA compliance
Backend	Node.js, Python, Kotlin (JHipster), REST, MQTT, Socket.IO, Firestore, MongoDB
AI	Claude Code, Codex, multi-agent orchestration, Cursor, MCP and skills, AI code review, OpenAI API, self-hosted LLMs (vLLM, llama.cpp)
Languages	Spanish and Catalan (native), English (C1)

EDUCATION

Multimedia Engineering studies — La Salle, Universitat Ramon Llull, Barcelona	2008–2014
Multimedia Studies — Universitat Oberta de Catalunya (UOC)	2013–2015